

Milena Rmus

Member of Technical Staff, Roundtable Technologies

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🌐 <https://milennacnlab.github.io/>

📍 San Francisco, CA

👤 Milena Rmus

Education

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| Aug 2019 – May 2024 | University of California, Berkeley
<i>PhD in Cognitive Science</i> |
| Aug 2014 – May 2018 | Brown University
<i>BS in Cognitive Neuroscience, Magna Cum Laude</i> |

Work Experience

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| Jan 2025 – Present | Roundtable Technologies <i>Member of Technical Staff</i> |
| May 2024 – Jan 2025 | Helmholtz Institute for Human-Centered AI Research <i>Scientist</i> |
| Aug 2019 – May 2024 | UC Berkeley <i>Graduate Student Researcher</i> |
| May 2022 – Aug 2022 | Lawrence Livermore National Lab <i>Data Science Intern</i> |
| Jun 2018 – Jun 2019 | Princeton University <i>Research Specialist</i> |

Publications

- Rmus, M., Hardy, M. D., Griffiths, T. L., & Agrawal, M. (2025). Process Matters more than Output for Distinguishing Humans from Machines. **Under review**. PDF
- Rmus, M., Jagadish, A. K., Mathony, M., Ludwig, T., & Schulz, E. (2025). Generating Computational Cognitive Models using Large Language Models. **NeurIPS**. PDF
- Binz, M., Jagadish, A.K., Rmus, M., & Schulz, E. (2025). Automated scientific minimization of regret. Advances in Neural Information Processing Systems **NeurIPS**. PDF
- Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., ... & Schulz, E. (2025). A foundation model to predict and capture human cognition. **Nature**. PDF
- Rmus, M., Eckstein, M. K. & Collins, A. G. E. (2025). Subgoals in Hierarchical Reinforcement Learning. **Under review**. PDF
- Rmus, M., Pan, T., Xia, L. & Collins, A. G. E. (2024). Artificial neural networks for model identification and parameter estimation in computational cognitive models. **PLOS Comp Bio**. PDF
- Rmus, M., He, M., Baribault, B., Walsh, E. G., Festa, E. K., Collins, A. G. E. & Nassar, M. R. (2023). Age-related differences in prefrontal glutamate are associated with increased working memory decay that gives the appearance of learning deficits. **eLife**. PDF
- Rmus, M., Zou, A. & Collins, A. G. E. (2023). Choice type impacts human reinforcement learning. **Journal of Cognitive Neuroscience**. PDF
- Rmus, M., Ritz, H., Hunter, L. E., Bornstein, A. M. & Shenhav, A. (2022). Humans can navigate complex graph structures acquired during latent learning. **Cognition**. PDF
- Rmus, M., McDougle, S. D. & Collins, A. G. E. (2021). The role of executive function in shaping reinforcement learning. **Current Opinion in Behavioral Sciences**. PDF

Peer-reviewed conference publications

- Rmus, M., Eckstein, M. K., & Collins, A. G. E. (2023). The role of subgoals in hierarchical reinforcement learning. Conference on Computational Cognitive Neuroscience. PDF

- Rmus, M., Xia, J., Collins, J. & Collins, A. G. E. (2022). Using Deep Learning tools for fitting Reinforcement Learning Models. Conference on Computational Cognitive Neuroscience. PDF
- Rmus, M. & Collins, A. G. E. (2020). What is a Choice in Reinforcement Learning? Proceedings of the 42nd Annual Meeting of the Cognitive Science Society. PDF

Conference presentations

- Rmus, M., & Jagadish, A. K. (2025). Guided Generation of Computational Cognitive Models (GeCCo). **Automated Scientific Discovery of Mind And Brain Workshop, Princeton University** [Invited Talk].
- Rmus, M., Jagadish, A. K., Mathony, M., Ludwig, T., & Schulz, E. (2025). Generating Computational Cognitive Models using Large Language Models. **Multidisciplinary Conference on Reinforcement Learning and Decision Making**. [Poster]
- Rmus, M. & Collins, A. G. E. (2023). Pressing a piano key, or playing a G note? Choice type impacts human reinforcement learning. **Society for Neuroscience**. [Talk]
- Rmus, M., Xia, J., Collins, J. & Collins, A. G. E. (2022). Using Deep Learning tools for fitting Reinforcement Learning Models. **Conference on Computational Cognitive Neuroscience**. [Poster]
- Rmus, M., He, M., Baribault, B., Festa, E. K., Collins, A. G. E. & Nassar, M. R. (2022). Reinforcement learning and working memory changes across lifespan: bridging cognition, computation and neuroscience. **Cognitive Neuroscience Society Annual meeting**. [Poster]
- Rmus, M. & Collins, A. G. E. (2020). What is a Choice in Reinforcement Learning? **Proceedings of the 42nd Annual Meeting of the Cognitive Science Society**. [Poster]
- Rmus, M. Mildner, J., Meyer, M., Hershfield, H., Waytz, A., & Tamir, D., Pattern dissimilarity distinguishes proximal and distal simulation in creative experts. (2019). **Social and Affective Neuroscience Society**. [Poster]
- Rmus M., Ritz, H., Hunter, L.E., Bornstein, & A.M., Shenhav, A. (2019). Model-based decision making is associated with structure inference ability. **Multidisciplinary Conference on Reinforcement Learning and Decision Making**. [Talk, presented by Harrison Ritz]
- Rmus M., Ritz, H., Hunter, L.E., Bornstein, A.M., & Shenhav, A. (2018). Model-based decision making is associated with structure inference ability. **Society for Neuroeconomics**. [Talk]

Teaching Experience

- PSYCH C123 Computational Models of Cognition, Graduate Teaching Assistant, UC Berkeley (2020, 2023)
- CLPS1590 – Visualizing Vision, Undergraduate Teaching Assistant, Brown University (2018)
- CLPS0220 – Making Decisions, Undergraduate Teaching Assistant, Brown University (2017)
- CLPS1700 – Abnormal Psychology, Undergraduate Teaching Assistant, Brown University (2017)
- CLPS0701 – Personality, Undergraduate Teaching Assistant, Brown University (2016)

Honors

- 2020 Graduate Remote Instruction Innovation Fellows Program (UC Berkeley)
- 2018 Teaching Assistant Kling Award (Brown University)
- 2018 Departmental Honors Award (Brown University)
- 2017 Brown Connect LINK Award (Brown University)

Technical Skills

Languages: Python (expert), Matlab (expert), JavaScript(fluent), R(fluent), SQL (prior experience), LaTeX(prior experience)

Frameworks and tools: Pandas, Keras, TensorFlow, PyTorch, NumPy, Scikit-Learn, jQuery, Matplotlib, Seaborn, ggplot2, Git, Notion, Trello, Adobe Illustrator, Adobe Photoshop, Procreate